

UW Quantum REU Solution

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Problem 1 (Distinguishing between two states). Let $n \in \mathbb{N}$, Suppose you are handed *one* of two n -qubit states $|\psi_0\rangle, |\psi_1\rangle$ uniformly at random. Assume that you know exactly what the two states are: you just don't know *which one* of the two you are given. Assume also that the two states are not identical (differing only by a global phase still counts as being identical here). In this question you will find the projective measurement $\{P, \mathbb{I} - P\}$ that maximizes the distinguishing success probability, i.e. it maximizes

$$\frac{1}{2}\Pr[0 | \psi_0] + \frac{1}{2}\Pr[1 | \psi_1] \quad (0.1)$$

where $\Pr[i | \psi_i]$ denotes the probability that the measurement outcome is i , given that the state was $|\psi_i\rangle$.

- (i) *Carefully* show that, without loss of generality, you can restrict your search to the 2-dimensional space spanned by $|\psi_0\rangle$ and $|\psi_1\rangle$, i.e. you can restrict yourself to consider measurements of the form

$$\{|u\rangle\langle u|, \mathbb{I} - |u\rangle\langle u|\} \quad (0.2)$$

where $|u\rangle$ lies in the span of $|\psi_0\rangle$ and $|\psi_1\rangle$

(You should try to make this proof as rigorous as possible – this question is more subtle than meets the eye!).

- (ii) Find the optimal measurement, and show that it is optimal.

Solution.

I. Notation:

- Given that $|\psi_0\rangle, |\psi_1\rangle$ are distinct in an n -qubit system (in particular they're normalized states with $\langle\psi_0|\psi_1\rangle = 0$), let $\mathcal{H} \cong \mathbb{C}^2$ be the 2-dimensional Hilbert Space for each qubit, then the n -qubit state is described by the space $\mathcal{H}^{\otimes n}$ (the tensor product of n copies of $\mathcal{H}$), which is finite-dimensional of dimension $d = 2^n$.
- Since $\mathcal{H}^{\otimes n}$ is a d -dimensional inner product space, hence one can specifically choose an orthonormal basis involving $|\psi_0\rangle, |\psi_1\rangle$, say $\{|\psi_0\rangle, |\psi_1\rangle, \dots, |\psi_{d-1}\rangle\} \subset \mathcal{H}^{\otimes n}$.
- Let $|u\rangle$ be a normalized state corresponding to the projection operator, we'll denote it as $|u\rangle = \sum_{k=0}^{d-1} a_k |\psi_k\rangle$, for $a_k \in \mathbb{C}$ that satisfies $\langle u|u\rangle = \sum_{k=0}^{d-1} |a_k|^2 = 1$. Then, the projection operator is of the following form:

$$|u\rangle\langle u| = \sum_{k=0}^{d-1} \sum_{l=0}^{d-1} a_k a_l^* |\psi_k\rangle\langle\psi_l| \quad (0.3)$$

The measurement outcome 0 will be read off from $\mathbb{I} - |u\rangle\langle u|$, while the measurement outcome 1 will be read off from $|u\rangle\langle u|$.

II. Calculating Distinguishing Success Probability:

To calculate $\Pr[0 \mid \psi_0], \Pr[1 \mid \psi_1]$, we have the following:

$$\begin{aligned}\Pr[0 \mid \psi_0] &= \langle \psi_0 | u \rangle \langle u | \psi_0 \rangle = \sum_{k=0}^{d-1} \sum_{l=0}^{d-1} a_k a_l^* \langle \psi_0 | \psi_k \rangle \langle \psi_l | \psi_0 \rangle \\ &= \sum_{k=0}^{d-1} \sum_{l=0}^{d-1} a_k a_l^* \delta_{0k} \delta_{0l} = a_0 a_0^* = |a_0|^2\end{aligned}\tag{0.4}$$

$$\begin{aligned}\Pr[1 \mid \psi_1] &= \langle \psi_1 | u \rangle \langle u | \psi_1 \rangle = \sum_{k=0}^{d-1} \sum_{l=0}^{d-1} a_k a_l^* \langle \psi_1 | \psi_k \rangle \langle \psi_l | \psi_1 \rangle \\ &= \sum_{k=0}^{d-1} \sum_{l=0}^{d-1} a_k a_l^* \delta_{1k} \delta_{1l} = a_1 a_1^* = |a_1|^2\end{aligned}\tag{0.5}$$

Hence, one has the distinguishing success probability with respect to $|u\rangle$ as:

$$\left(\frac{1}{2} \Pr[0 \mid \psi_0] + \frac{1}{2} \Pr[1 \mid \psi_1] \right)_{|u\rangle} = \frac{1}{2} (|a_0|^2 + |a_1|^2)\tag{0.6}$$

Where, a_0, a_1 are the coefficients of $|\psi_0\rangle, |\psi_1\rangle$ in $|u\rangle$ respectively (and this formula is true for arbitrary $|u\rangle \in \mathcal{H}^{\otimes n}$).

III. Solutions:

- (i) To show that one only needs to check $|u\rangle \in \text{span}\{|\psi_0\rangle, |\psi_1\rangle\}$, we'll show that if $|u\rangle \notin \text{span}\{|\psi_0\rangle, |\psi_1\rangle\}$, there exists some other normalized state $|u'\rangle \in \text{span}\{|\psi_0\rangle, |\psi_1\rangle\}$, such that $\{|u'\rangle, \mathbb{I} - |u'\rangle\langle u'|\}$ generates a higher distinguishing success probability.

Suppose $|u\rangle \notin \text{span}\{|\psi_0\rangle, |\psi_1\rangle\}$, this implies some $a_2, \dots, a_{d-1} \in \mathbb{C}$ are nonzero, say $i_1, \dots, i_k \in \{2, \dots, d-1\}$ are all distinct indices (other than indices 0, 1) such that each $a_{i_j} \neq 0$. This implies the following:

$$1 = \langle u | u \rangle = \sum_{j=0}^{d-1} |a_j|^2 = |a_0|^2 + |a_1|^2 + \sum_{j=1}^k |a_{i_j}|^2\tag{0.7}$$

since the remaining coefficients are 0).

Then, consider the state $|u'\rangle = \left(\sqrt{|a_0|^2 + \sum_{j=1}^k |a_{i_j}|^2} \right) |\psi_0\rangle + a_1 |\psi_1\rangle$. Which, $|u'\rangle \in \text{span}\{|\psi_0\rangle, |\psi_1\rangle\}$, and moreover it satisfies the following:

$$\langle u' | u' \rangle = \left| \sqrt{|a_0|^2 + \sum_{j=1}^k |a_{i_j}|^2} \right|^2 + |a_1|^2 = |a_0|^2 + |a_1|^2 + \sum_{j=1}^k |a_{i_j}|^2 = 1\tag{0.8}$$

This shows that $|u'\rangle$ is perfectly normalized. On the other hand, notice that the distinguishing success probability of $|u'\rangle$ is given as follow:

$$\begin{aligned}
\left(\frac{1}{2} \Pr[0 \mid \psi_0] + \frac{1}{2} \Pr[1 \mid \psi_1]\right)_{|u'\rangle} &= \frac{1}{2} + \frac{1}{2} \left(\left| \sqrt{|a_0|^2 + \sum_{j=1}^k |a_{i_j}|^2} \right|^2 - |a_1|^2 \right) \\
&= \frac{1}{2} + \frac{1}{2} \left(|a_0|^2 + \sum_{j=1}^k |a_{i_j}|^2 - |a_1|^2 \right) \\
&> \frac{1}{2} + \frac{1}{2} (|a_0|^2 - |a_1|^2) \\
&= \left(\frac{1}{2} \Pr[0 \mid \psi_0] + \frac{1}{2} \Pr[1 \mid \psi_1]\right)_{|u\rangle}
\end{aligned} \tag{0.9}$$

This strict inequality holds because each $a_{i_j} \neq 0$, showing $|a_{i_j}|^2 > 0$. Which, this demonstrates that $|u'\rangle \in \text{span}\{|\psi_0\rangle, |\psi_1\rangle\}$ generates a higher distinguishing success probability than $|u\rangle$, providing that $|u\rangle \notin \text{span}\{|\psi_0\rangle, |\psi_1\rangle\}$.

hence, we conclude that any $|u\rangle \notin \text{span}\{|\psi_0\rangle, |\psi_1\rangle\}$ CANNOT maximize the distinguishing success probability (since one can always choose some other normalized state with higher probability). Which, one only needs to check $|u\rangle$ that lies in the span of $|\psi_0\rangle, |\psi_1\rangle$.

- (ii) Suppose $|u\rangle = x|\psi_0\rangle + y|\psi_1\rangle$ is a normalized state maximizing the distinguishing success probability (which, $\langle u|u\rangle = |x|^2 + |y|^2 = 1$, showing $|x|, |y| \in [0, 1]$). WLOG, by multiplying with suitable global phase, one can assume $x \in [0, 1]$.

Then, notice that $|y|^2 = 1 - |x|^2 = 1 - x^2$, which shows the following for the distinguishing success probability:

$$\begin{aligned}
\left(\frac{1}{2} \Pr[0 \mid \psi_0] + \frac{1}{2} \Pr[1 \mid \psi_1]\right)_{|u\rangle} &= \frac{1}{2} + \frac{1}{2} (|x|^2 - |y|^2) \\
&= \frac{1}{2} + \frac{1}{2} (x^2 - (1 - x^2)) \\
&= x^2
\end{aligned} \tag{0.10}$$